

PRANAY KUMAR

AI Researcher | ML/DL | Computer Vision

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PROFILE

I'm a 4th-year Integrated M.Tech student who genuinely enjoys digging into how AI systems break — and how to make them better. Most of my research has revolved around adversarial robustness and the security of Vision-Language Models and DNNs, work I've had the chance to do at IISER Bhopal and IIT Bhubaneswar, which got me thinking a lot about how these models behave in the real world. Currently exploring VLMs and Generative Models — always chasing the next hard problem where trustworthy AI still falls short.

EDUCATION

Integrated M.Tech in INFORMATION TECHNOLOGY

Aug 2023 – Present

International Institute of Professional Studies, DAVV, Indore | CGPA: 7.57 / 10

Senior Secondary (12th Board)

Apr 2023

Score: 80.4 / 100

RESEARCH & PROJECTS

Research Intern

Present

IISER Bhopal

- Working on security and robustness of Vision-Language Models (VLMs)

Research Intern

Dec 2025 – Feb 2026

IIT Bhubaneswar (Remotely)

- Worked on an LLM-based Trip Planner in collaboration with Microsoft India
- Designed and created structured trip plans used for model evaluation and benchmarking

Research Intern

May 2025 – Jul 2025

IISER Bhopal

- Conducted experiments based on the ICLR 2025 paper: Democratic Training Against Universal Adversarial Perturbations
- Fine-tuned deep learning models to enhance adversarial robustness
- Implemented entropy regularization and inter-layer agreement techniques for improved adversarial defense

SKILLS

Programming: Python, C++, MySQL

Frameworks & Libraries: PyTorch, NumPy, Pandas, Matplotlib

AI/ML Domains: Deep Learning, Computer Vision, Transformers, LLMs, Vision-Language Models (VLMs), Adversarial Robustness

Other: Generative AI, Social Network Analysis, Data Structures

NPTEL COURSEWORK

- Deep Learning for Computer Vision
- Introduction to Large Language Models (LLMs)

- Programming with Generative AI
- Social Network Analysis

PRESENTATIONS & WORKSHOPS

Participant & Presenter	<i>July 2025</i>
• Presented research on adversarial robustness and democratic training in AI/ML systems • Participated in workshops on AI ethics, deep learning, LLMs, and interpretability	
Workshop	<i>June 2025</i>
• Explored data forgetting techniques and ethical model design for generative AI systems	
Symposium	<i>March 2025</i>
• Attended sessions on AI-driven diagnostics, medical imaging, and healthcare innovation	

LANGUAGES

English (Professional), Hindi (Native)